Evolving a microstrip pair from Maxwell's equations

Can a GA find a 50 Ω, low-crosstalk trace pair when every candidate's L and C come from a field solve? — milestone M9i, CLAUDE.md §3c

kronspice simulation report · 2026-11-07 · population 20, 12 generations, seed 1

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Summary

Two parallel microstrip traces on FR-4 are evolved for a 50 Ω characteristic impedance and near-end crosstalk (NEXT) no worse than −30 dB at 100 MHz, on the least board width. Nothing about the traces is looked up: for every candidate the per-unit-length inductance and capacitance matrices are extracted from the cross-section by the quasi-static field solver of milestone M9i, the pair becomes a coupled-line ladder in the circuit, and an AC analysis measures the crosstalk onto the quiet trace.

- **Evolved design (218 evaluations):** width w = 2.766 mm, gap s = 3.020 mm, board width 8.55 mm.
- **It meets the specification:** Z_0 = 49.82 Ω and NEXT = −33.1 dB at 100 MHz.
- **Extracted line parameters:** ε_{eff} = 3.257, L_{11} = 301 nH/m, C_{11} = 121.3 pF/m, coupling K = L_{12}/L_{11} = 0.0857.
- **Coupling falls with the gap** for a 3 mm trace: K = 0.287, 0.128, 0.030 and NEXT = −20.8, −29.1, −43.3 dB at s = 0.5, 2, 6 mm.
- **The field solver is validated** against coax (+0.07 %), Cohn's stripline (−0.006 % extrapolated), the two-wire line (0.1 %) and Hammerstad–Jensen microstrip (Z_0 −1.8 %, ε_{eff} −0.4 %).

Background

CLAUDE.md §3c asks that circuit elements be traceable to Maxwell's equations, and that *em/* supply lumped parameters from geometry so evolved circuits can include physical layout and parasitic coupling. Milestone M9i (*doc/M9i.md*) builds that extraction; this example is its demonstration: a genetic algorithm that evolves physical trace geometry, scored by a circuit simulation whose element values come from a field solve. No hypothesis is involved.

The model

- **Extraction** (*em.Section*): strips of zero thickness on top of a 1.6 mm substrate, ε_r = 4.4, air above, inside a box twelve substrate thicknesses wide. The equation div(ε grad φ) = 0 is solved with each conductor driven in turn, giving the 2×2 capacitance matrix C. The same solve with ε = 1 gives C_0, and $L = \mu_0\varepsilon_0 C_0^{-1}$ (the TEM relation).
- **Circuit** (*em.CoupledLadder*): the 10 cm pair as 20 sections of the coupled telegrapher's equations, mutual inductance as a K element and mutual capacitance as a capacitor between the traces. The aggressor trace is driven through Z_0, and both traces are terminated in Z_0 at both ends.
- **Measurement:** NEXT = |v(v_in)/v(a_in)| in dB from a one-point AC analysis at 100 MHz (v_in is the near end of the quiet trace, a_in the near end of the driven one).
- **Genome:** trace width w (0.3–6 mm) and gap s (0.2–10 mm).
- **Objectives (weighted sum):** $|Z_0 - 50|/50$; crosstalk above −30 dB; and 0.01 × board width (2w + s) in units of 10 mm. Each candidate's extraction is memoized across the objectives.

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Results

Evolved pair

Run	Evals	w	s	Board	Z_0	ε_{eff}	$K = L_{12}/L_{11}$	NEXT at 100 MHz
Default (pop 20, 12 gens, seed 1)	218	2.766 mm	3.020 mm	8.55 mm	49.82 Ω	3.257	0.0857	−33.1 dB
Shorter (doc/M9i.md, -gens 8 -pop 16)	114	2.769 mm	8.913 mm	14.45 mm	49.86 Ω	3.281	0.0130	−50.8 dB

The default run also reports L_{11} = 301 nH/m and C_{11} = 121.3 pF/m. Both runs land on almost the same width, which sets Z_0. The shorter run settled on a wider gap; the board-width weight is small, so either design satisfies the specification.

Coupling against gap

For a 3 mm trace (checked by the example's test):

Gap s	$K = L_{12}/L_{11}$	NEXT at 100 MHz
0.5 mm	0.287	−20.8 dB
2 mm	0.128	−29.1 dB
6 mm	0.030	−43.3 dB

Validation

Field extraction (doc/M9i.md, em/extract_test.go)

Case	Reference	Result
Coax, ε_r = 2.1	$2\pi\varepsilon/\ln(b/a)$	+0.07 % (C, L, Z_0)
Stripline, zero thickness	Cohn 1954, exact K(k)/K(k′)	−0.006 % after Richardson extrapolation (b/40, b/80)
Two-wire line	$\pi\varepsilon_0/\operatorname{acosh}(D/2a)$	0.1 % in a 120 mm box
Microstrip, FR-4, w = 3 mm	Hammerstad–Jensen 1980	Z_0 −1.8 %, ε_{eff} −0.4 %

The telegrapher ladder is checked in *em/tline_test.go*: a 2 m microstrip-like line gives Z_0 51.85 Ω and a transient delay of 9.67 ns (0.690 c), matching $\sqrt{(L/C)}$ and $\sqrt{(LC)}$.

Kron's torn solution against MNA

The exported w = 3 mm, s = 2 mm pair (185 elements, 83 nodes; *export/pcb_layout_pair_w3_s2_results.md*) is solved at 100 MHz both by full MNA and by Kron's diakoptics, torn automatically into 2 pieces of 2 and 162 unknowns joined by 1 link (the source resistor RS, carrying 0.0104559 A at phase 0.01°). The largest node-voltage difference is 1.2×10^{-15} of the largest node voltage, and the power taken by the R, L, C branches is 0.0104559 at phase −0.01° in both (power invariant under tearing, Roth 1959 §5). In that solve the quiet trace's near end is 0.0176376 V at phase 68.20° against 0.50019 V at phase −0.01° on the driven trace.

Determinism

The GA test runs without a per-evaluation timeout and fails if any candidate timed out, so the evolved result is the same on every machine (CLAUDE.md §1 goal 5).

Limits of the model

- Strips have zero thickness.
- Quasi-static (TEM) only: dispersion and radiation are left out (*em/limits.go* applies).
- The ladder's own λ/10 rule sets the section count (20 sections for 10 cm).
- Crosstalk is measured at a single frequency, 100 MHz.

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Reproducing the results

```
go run ./examples/pcb_layout [-gens 12] [-pop 20]
go test ./examples/pcb_layout          # -short skips the GA run
KRONSPICE_EXPORT=1 go test ./examples/pcb_layout -run TestExport
```

Sources: C. R. Paul, *Analysis of Multiconductor Transmission Lines*, 2nd ed. (2008); E. Hammerstad, Ø. Jensen, "Accurate models for microstrip computer-aided design," *IEEE MTT-S* (1980); doc/M9i.md.

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